Characterizing Insecure Error Distributions For Various RLWE Problems

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April 13, 2021

Outline

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Introduction

- Most cryptosystems that have been in use over the past few decades are threatened by their lack of quantum-level security.
- It is predicted that quantum computers will be able to attack classical-level cryptosystems on a far shorter timeline than ordinary classical computers.
- This has led government agencies (led by NIST) to search for a new quantum-secure cryptosystem.
- We want a cryptosystem that is **quantum-safe** and **fully-homomorphic**.

FHE, Defined Mathematically

- Suppose we have a set of **sensitive** data in a database that we want to store securely on a server through the cloud.
- Let D denote our unencrypted data set and E denote our encrypted data set.
- Define a map $\varphi : D \to E$.
- We want to be able to access data and perform computations over φ:
  1. $\varphi(a + b) = \varphi(a) + \varphi(b)$
  2. $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$
- ...without exhausting **resources** or compromising **security**
- If $+$ and $\cdot$ are respected in φ, we have **fully homomorphic encryption** (FHE).

Introduction to RLWE

- We're interested in a class of problems known as **Ring Learning with Errors** (RLWE), which involves hiding secret vectors in a multidimensional lattice (Remark: such a space is a $\mathbb{Z}$-module, and the vectors have $\mathbb{R}$ entries).
- We also introduce errors into the values we send (without errors, we can find our secret using techniques from linear algebra).
- For simplicity, we will only discuss a specific variant of RLWE known as **Polynomial Learning with Errors**, which hides a polynomial secret instead of a vector over the polynomial ring $(\mathbb{Z}/q\mathbb{Z})[x]$.

PLWE Construction

How to construct a PLWE instance:

- Fix a monic irreducible polynomial $f \in \mathbb{Z}[x]$ of degree n.
- Fix a prime integer modulus q.
- Let $R = \mathbb{Z}[x]/(f(x))$.
- Compute the quotient ring $R_q = R/qR$
- If we have a polynomial in R,
  $a_0 + a_1 x + a_2 x^2 + \ldots + a_{n-1} x^{n-1}$, $a_i \in \mathbb{Z}$, then there is a polynomial $A(x) \in R_q$ where
  $A(x) = \overline{a_0} + \overline{a_1} x + \ldots + \overline{a_{n-1}} x^{n-1}$ and $\overline{a_i} \equiv a_i \mod q$.

Remark: We choose f such that the ring of integers of the field $K = \mathbb{Q}[x]/(f(x))$ is monogenic so that $\mathcal{O}_K = \mathbb{Z}[x]/(f(x))$.

The Error Distribution

- For security, we will hide our secret by adding errors in the form of polynomials with coefficients drawn randomly from an **error distribution**, D_σ (where σ is the width of the error distribution).
- Each error takes the form

$$E(x) = \overline{e_0} + \overline{e_1}x + \ldots + \overline{e_{n-1}}x^{n-1},$$

where each $\overline{e_i} \in \mathbb{Z}/q\mathbb{Z}$ is chosen independently according to a discrete Gaussian distribution. (The distribution D_σ is the joint distribution of the $\overline{e_i}$ terms).

PLWE Construction, cont.

The basics of PLWE encryption:

- Pick a secret $S \in R_q$
- For $j = 1, \ldots, m$:
  - Pick A_j uniformly at random from R_q
  - Pick E_j at random from the Gaussian distribution (it is important that the coefficients of E_j are small relative to A_j)
- Create new polynomials $B_j = (A_j \cdot S) + E_j$
- This gives us the cryptosystem's **public data**: a set of pairs $\{(A_1, B_1), (A_2, B_2), \ldots, (A_m, B_m)\}$

Note: We omit the specific details of encryption and decryption for this presentation.

The RLWE Problem

The (search) RLWE computational problem involves discovering a secret $S \in R_q$ given an arbitrary number of samples (A_i, B_i) from $R_q \times R_q$ where $B_i = (A_i \cdot S) + E_i$ as before.

PLWE Attack Homomorphism

There exists a map:

$$R_q \to \mathbb{Z}/q\mathbb{Z} = \mathbb{F}_q$$
$$A \mapsto \overline{A}$$

that is a ring homomorphism.

Remark: This map is reduction modulo $\mathfrak{q}$ for $\mathfrak{q}$ above q in K; we choose q so that the primes above it have residue field $\mathbb{F}_q$.

PLWE Attack, cont.

This means that if

$$B = (A \cdot S) + E$$

then

$$\overline{B} = (\overline{A} \cdot \overline{S}) + \overline{E}$$

Guessing the Secret

Because $\mathbb{F}_q$ is much smaller than R_q (enough to where its elements can be enumerated in a reasonable amount of time), the attacker can try to directly guess $\overline{S} \in \mathbb{F}_q$ by trial and error.

Guessing the Secret, cont.

If a guess $g \neq \overline{S}$, then

$$\begin{aligned}
&\overline{B_j} - (g \cdot \overline{A_j}) \\
&= (\overline{A_j} \cdot \overline{S}) + \overline{E_j} - (g \cdot \overline{A_j}) \\
&= \overline{A_j}(\overline{S} - g) + \overline{E_j}
\end{aligned}$$

and this quantity is a random number in $\mathbb{F}_q$. These numbers would be uniformly distributed, which would indicate an incorrect guess.

Guessing the Secret, cont.

However, if $g = \overline{S}$, then

$$\begin{aligned}
&\overline{B_j} - (g \cdot \overline{A_j}) \\
&= (\overline{A_j} \cdot \overline{S}) + \overline{E_j} - (g \cdot \overline{A_j}) \\
&= \overline{A_j}(\overline{S} - g) + \overline{E_j} \\
&= \overline{A_j}(\overline{S} - \overline{S}) + \overline{E_j} \\
&= \overline{E_j}
\end{aligned}$$

Under certain circumstances, the distribution of the values $\overline{E_j}$ is not uniform - this indicates a correct guess.

The Point

The goal of this research project is to determine these circumstances - what affects the $\overline{E_j}$ distribution's uniformity or lack thereof.

Introduction to Research Phase

Peikert[1] gives conditions for the size of the prime modulus q and the width of the error distribution σ to ensure that the scheme is secure. However, our research suggests that Peikert's conditions for security in a RLWE instance are sufficient, but not necessary. We find that there is in fact a trade-off, where the width can be narrower if the prime is smaller.

[1] Chris Peikert. "How (Not) to Instantiate Ring-LWE". In: *Lecture Notes in Computer Science Security and Cryptography for Networks* (2016), pp. 411–430. DOI: 10.1007/978-3-319-44618-9_22.

Methodology

For this project, we adapted code from Chen, Lauter, and Stange[2]. After some modifications, the program implements the specific attack outlined above and uses a Chi-Squared statistical test to determine the error distribution uniformity. We proceeded to run tests on a number of sets of parameters - consisting of a representative polynomial f for some number field, a prime modulus q, and a width distribution (σ in this presentation, also referred to in the below graphs as s) and found several particularly weak RLWE instances.

[2] Hao Chen, Kristin E. Lauter, and Katherine E. Stange. "Attacks on the Search-RLWE problem with small errors". In: *CoRR* abs/1710.03739 (2017). arXiv: 1710.03739. URL: http://arxiv.org/abs/1710.03739.

Result Analysis

The next set of slides shows graphs of some data we collected from running this program on various parameters.

Important: A "successful attack" in this case indicates weakness, whereas a "failure" indicates the instance's defense against the attack.

Results I

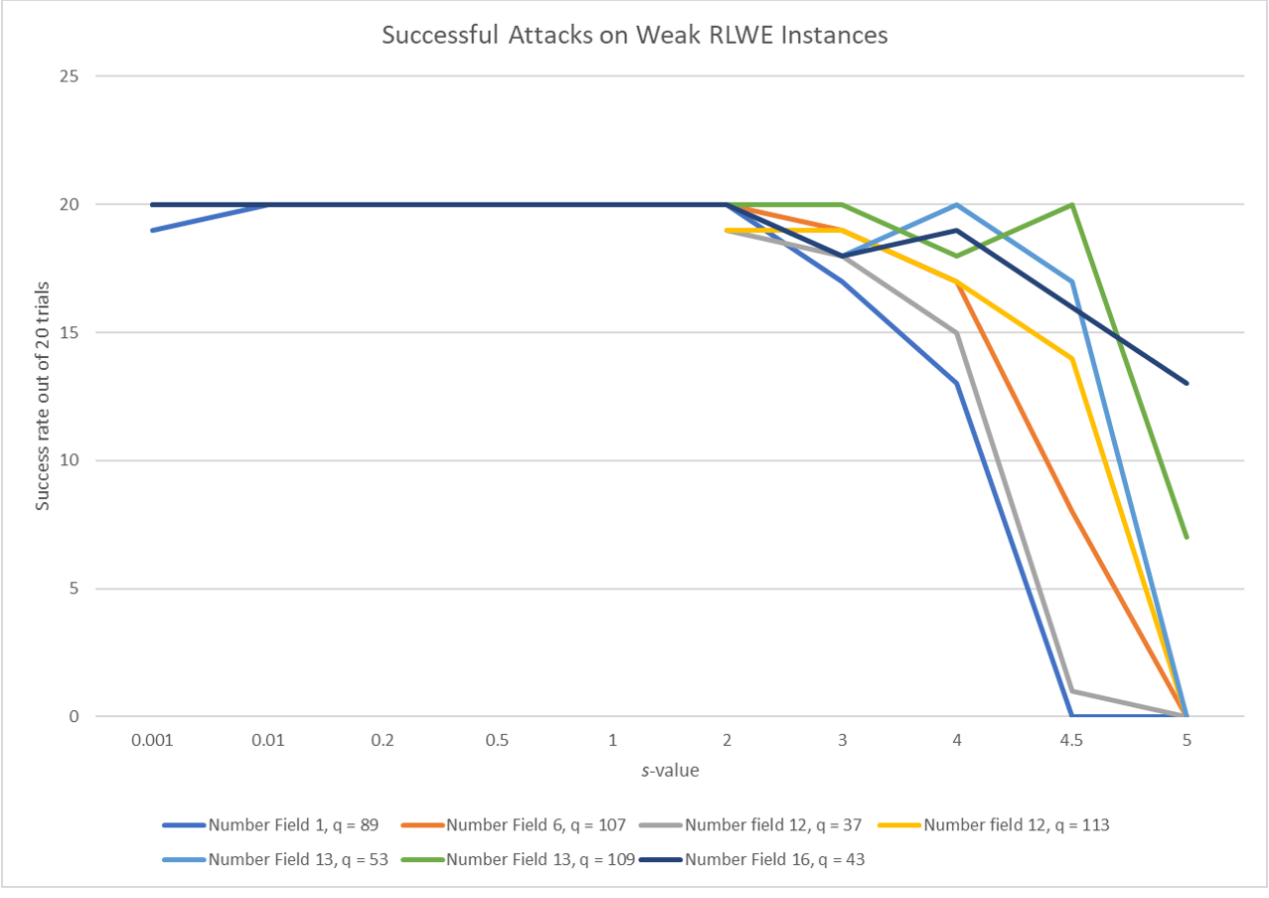


Figure 1: The success rate of the RLWE attack program on five polynomial/prime modulus combinations, with varying σ-value over the x-axis.

Results II

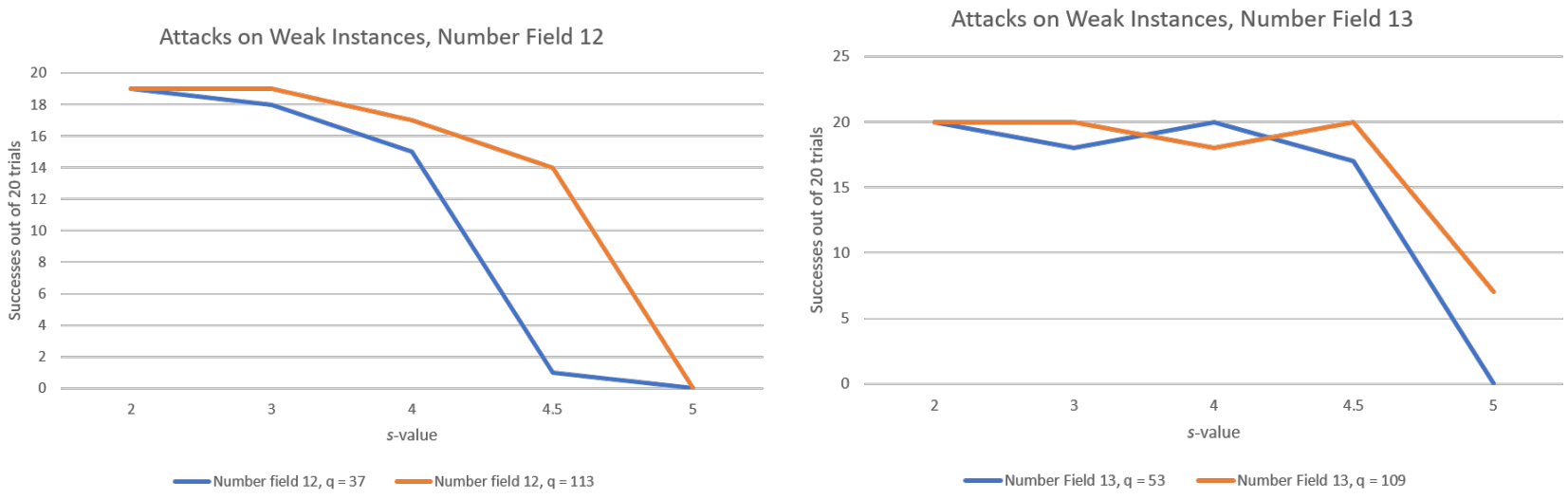


Figure 2: The two weak RLWE instances each with the polynomials for number fields 12 and 13. This graph shows how q renders the instance vulnerable to the attack longer as σ decreases.

Results III

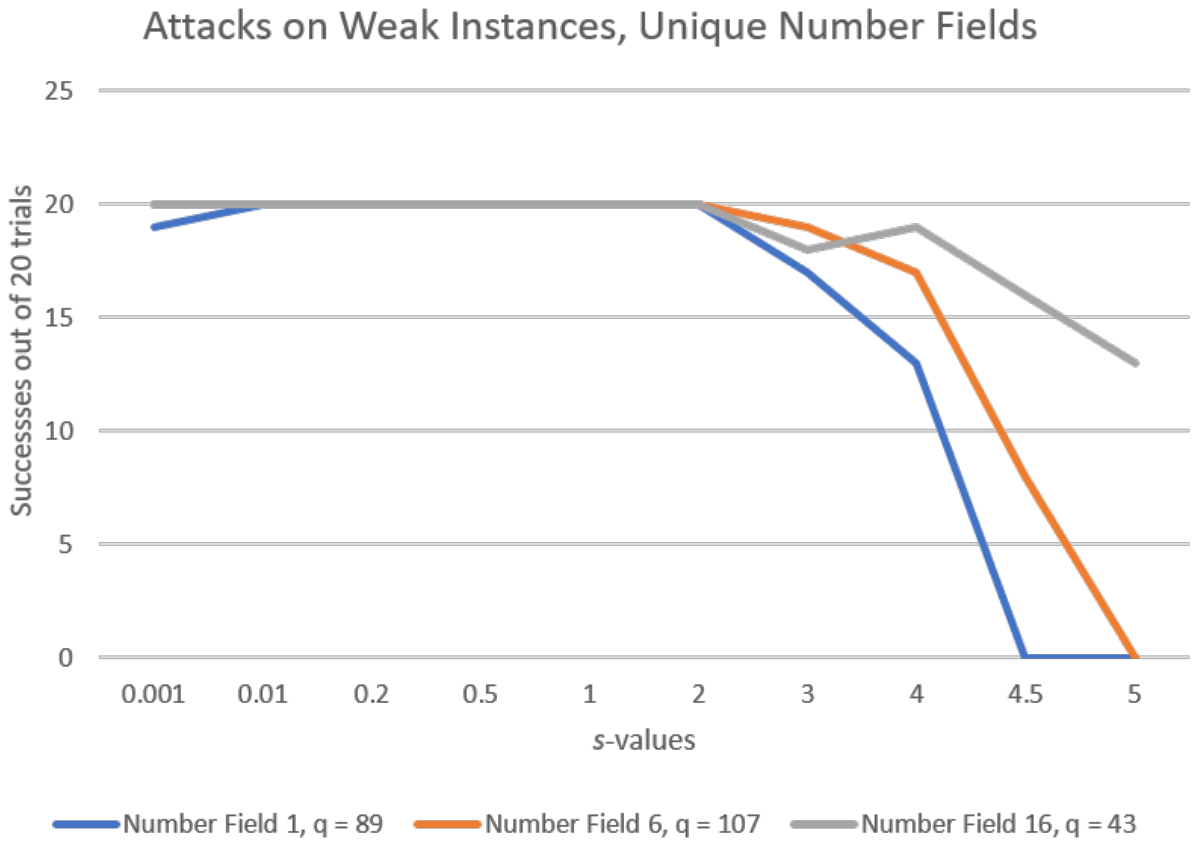


Figure 3: The weak RLWE instances with unique number fields. This graph shows how the size of the polynomial's degree negatively affects the instance's ability to withstand the attack.

Further Considerations

- The qualities of the number field that affect a RLWE instance's weakness
- Testing more polynomial, prime, σ values to make a conjecture
- Defining the σ-point of security in particularly weak instances
- What about robust instances (secure even when $\sigma < 0.1$)? Where are they insecure?

Q & A

Questions?